(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: ETMA-102 ***Subject: Applied Mathematics-II***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No1 which is compulsory. Select one question from each unit.

Q1).

a) If $u = e^{xyz}$, show that $\frac{\partial^3 u}{\partial x\,\partial y\,\partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$. [4]

b) Form a partial differential equation of $z = (x^2 + a^2)(y^2 + b^2)$. [4]

c) Find Laplace transform of $f(t) = te^{-4t}\sin 3t$. [4]

d) Prove that the function sinhz is analytic and find its derivative. [4]

e) Check whether the vector field $F = 2xyz^3i + x^2z^3j + 3x^2yz^2k$ is irrotational. [4]

f) Evaluate $\int_0^1 \int_{e^x}^{e} \frac{1}{logy}\,dydx$, *by changing the order of integration.* [5]

UNIT - I

Q2.

a) If $x + y + z - u = 0,\ y + z - uv = 0,\ z - uvw = 0,$ show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$. [6]

b) Examine the function $f(x, y) = x^3 + y^3 - 12x - 3y + 20$ for extremum. [6.5]

Q3.

a) Find the maximum and minimum distances of the point $(1,2,-1)$ from the sphere $x^2 + y^2 + z^2 = 24$. [6.5]

b) If $V = r^m$, and $r^2 = x^2 + y^2 + z^2$, show that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = m(m + 1)r^{m-2}$. [6]

UNIT-II

Q4.

a) Find Laplace transform of $\int_0^t \int_0^t \int_0^t (tsint)dtdtdt$. [6]

b) State convolution theorem and using it find inverse Laplace transform of $\frac{1}{s^2(s^2+a^2)}$ [6.5]

Q5.

a) Find Laplace transform of $f(t) = \frac{cos2t - cos3t}{t}$. [6]

b) Using Laplace transform solve: $y''' + 2y'' - y' - 2y = 0$, given $y(0) = 1,\ y'(0) = 2,$ $y''(0) = 2$. [6.5]

UNIT - III

Q6.

a) Find the bilinear transformation which maps the points $z = 1, -i, -1$ into the points $w = i, 0, -i$ [6]

b) Evaluate the integral $\int_0^{2\pi} \frac{d\theta}{1 - 2asin\theta + a^2}$, $0 < a < 1$. [6.5]

P.T.O.